Creating our first TypeScript application

Let's start by ensuring that you have all the necessary tools installed. By now, you should have Visual Studio Code, Node.js, npm, and TypeScript Compiler (TSC) installed on your computer. If not, please follow the installation guides provided attached to this document.

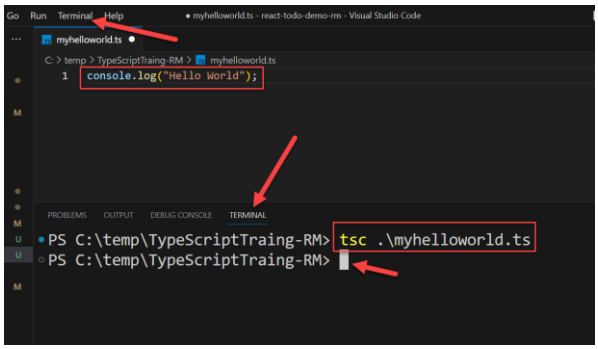
1. Visual Studio Code
2. node
3. npm
4. tsc

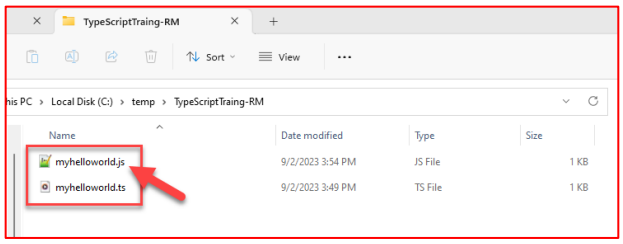
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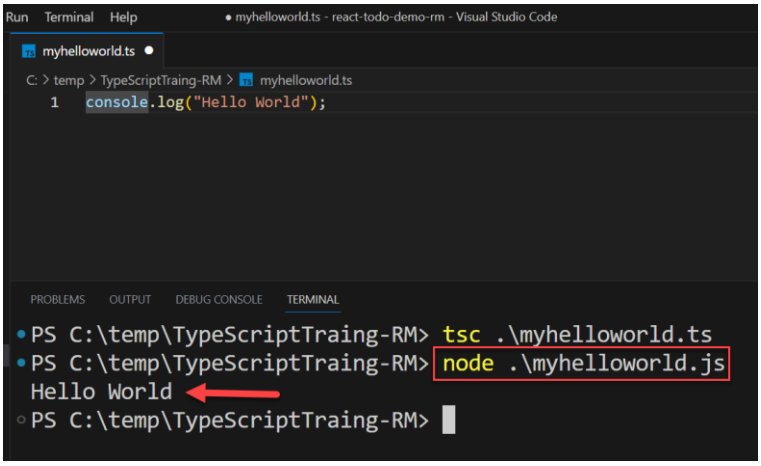
Now, let's organize our workspace. We're going to create a new folder to store all the code for this training series. I'll name it 'TypeScriptTraining', but feel free to choose any name you prefer. Next, we'll open Visual Studio Code, which we'll refer to as the IDE (Integrated Development Environment) from now on. For demonstration, we're using the default settings. We'll then open the 'TypeScriptTraining' folder in the IDE.

Let's create a subfolder under 'TypeScriptTraining' called '01_HelloWorld' and a new file within it called 'MyHelloWorld.ts'. Remember, `'.ts'` is the file extension for TypeScript files. We'll start with a simple program that just logs 'Hello World' to the console. Notice how IntelliSense code completion assists us as we type.

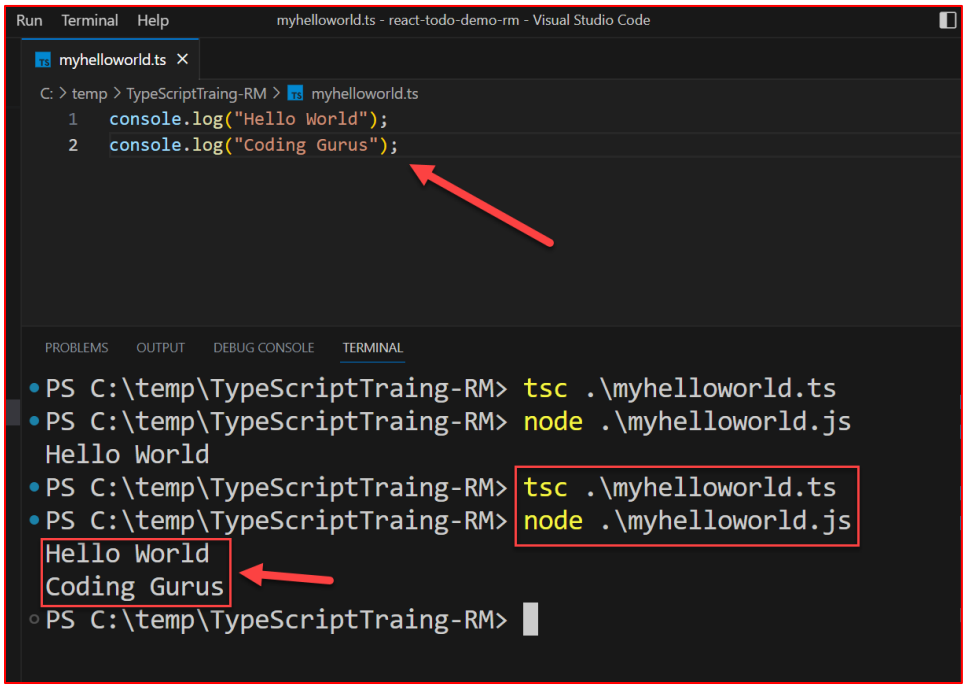
Now, let's compile and run our code. I'll open a new terminal in the IDE, navigate to the '01_HelloWorld' directory, and check its contents to confirm that `'MyHelloWorld.ts'` is there. To compile the code, I'll use the `'tsc'` command followed by the filename. This will generate a JavaScript file, `'MyHelloWorld.js'`, which we can then run using the 'node' command. And there we have it, our 'Hello World' output!







Let's add some more code to our TypeScript file. I'll add another `'console.log'` statement and concatenate some strings. After saving the changes, I'll clear the terminal, recompile the TypeScript file, and run the newly generated JavaScript file. The new output should reflect our changes.



A quick note of caution: always edit the TypeScript files, not the generated JavaScript files. The JavaScript files are regenerated every time we compile the TypeScript code, so any changes made directly to the JavaScript files will be lost.